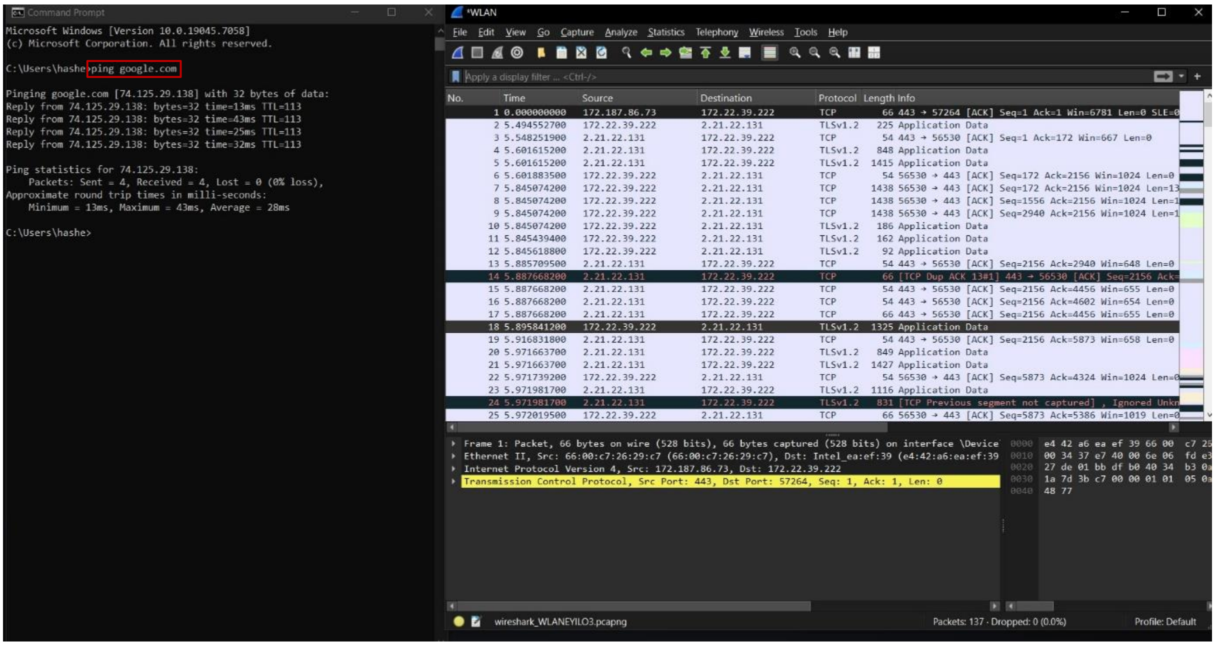
ICMP Traffic Analysis Using Wireshark

Introduction

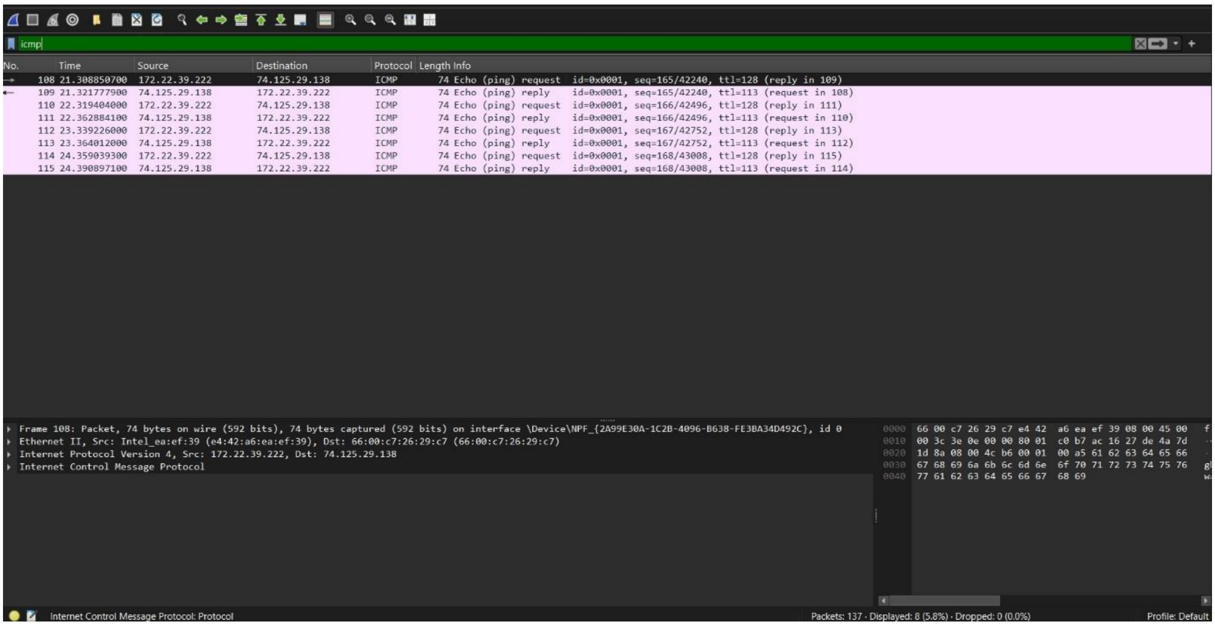
As a network engineer assigned to investigate potential ICMP-related attacks, I used Wireshark to capture and analyze ICMP traffic on the network. My investigation involved two main tests: a ping test and a traceroute test on google.com. These tests helped me understand how ICMP packets behave and identify any unusual patterns that could indicate a security threat (Ghosh, 2021).

Part 1: ICMP Ping Analysis

To start the investigation, I opened Wireshark and selected the WLAN interface. Then I opened the Command Prompt and ran a ping test on google.com. The ping was successful, with 4 packets sent and 4 received, showing 0% packet loss.

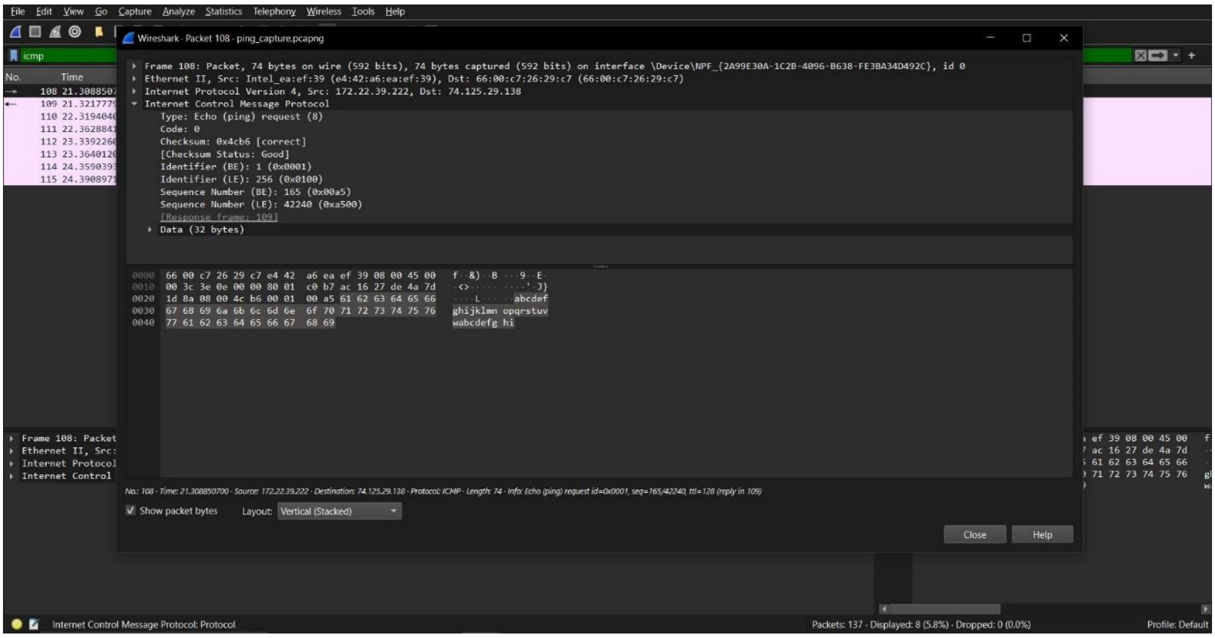


After applying the ICMP filter in Wireshark, I could clearly see the ping request and reply packets.



Question 1a: Protocol and Protocol Number

The first ICMP request packet I found was Packet 108. The protocol is the Internet Control Message Protocol, known as ICMP, and its protocol number is 1. This protocol operates at the Network Layer and is mainly used for diagnostic and error-reporting purposes (Ghosh, 2021).

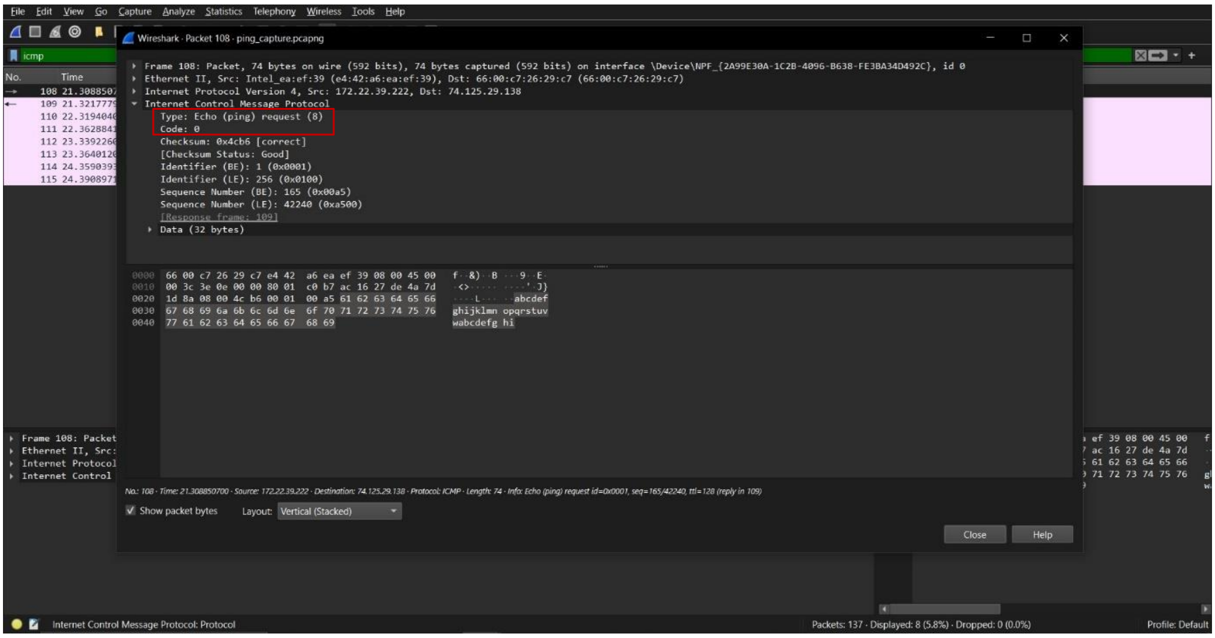


Question 1b: Type and Code - ICMP Request

Type: 8 = Echo (ping) request

Code: 0

A Type value of 8 means the packet is an Echo Request, which is sent from my PC to Google to test the connection. The Code value of 0 is the standard code for this type of message (Ghosh, 2021).



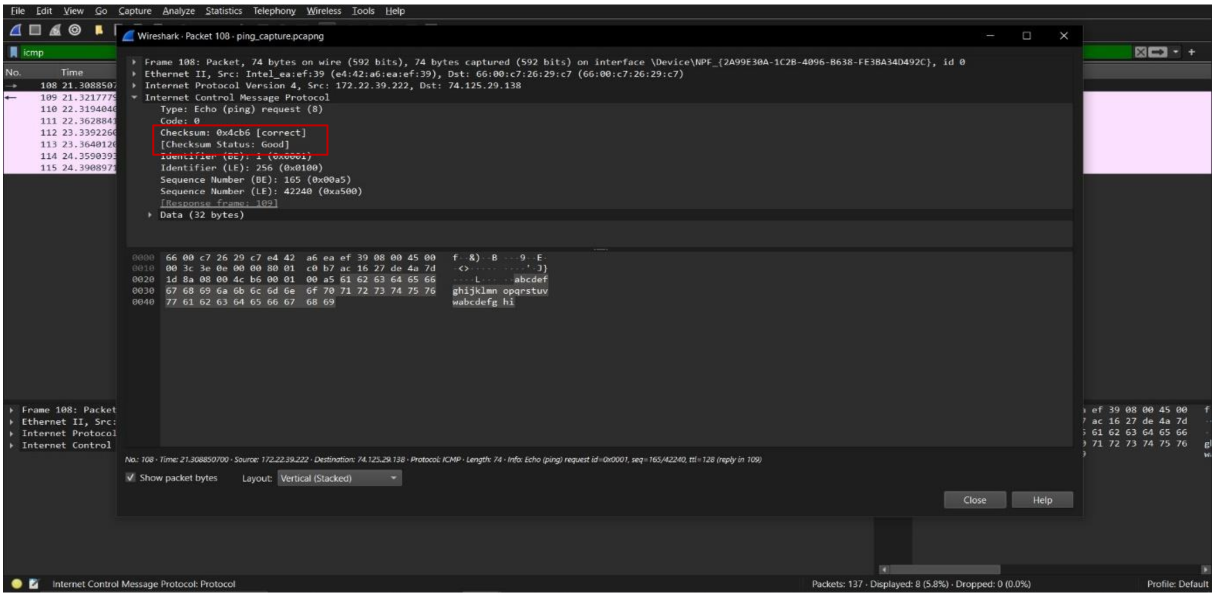
Question 1c: Checksum and Status

Checksum: 0x4cb6

Status: Correct

Checksum Status: Good

The checksum was verified by Wireshark and confirmed as correct, meaning the packet arrived without any corruption or errors (Ghosh, 2021).



Part 2: ICMP Reply Packet Analysis

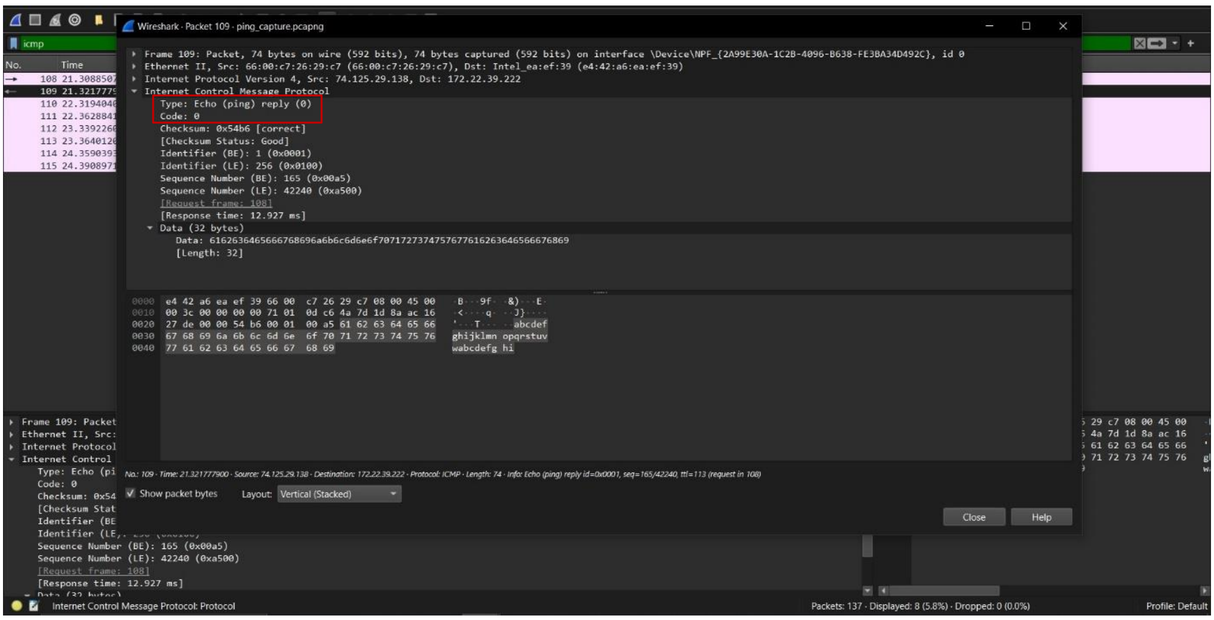
The reply packet was Packet 109, sent from Google's server back to my PC.

Question 2d: Type and Code - ICMP Reply

Type: 0 = Echo (ping) reply

Code: 0

A Type value of 0 means the packet is an Echo Reply, confirming that Google received my request and responded successfully (Ghosh, 2021).

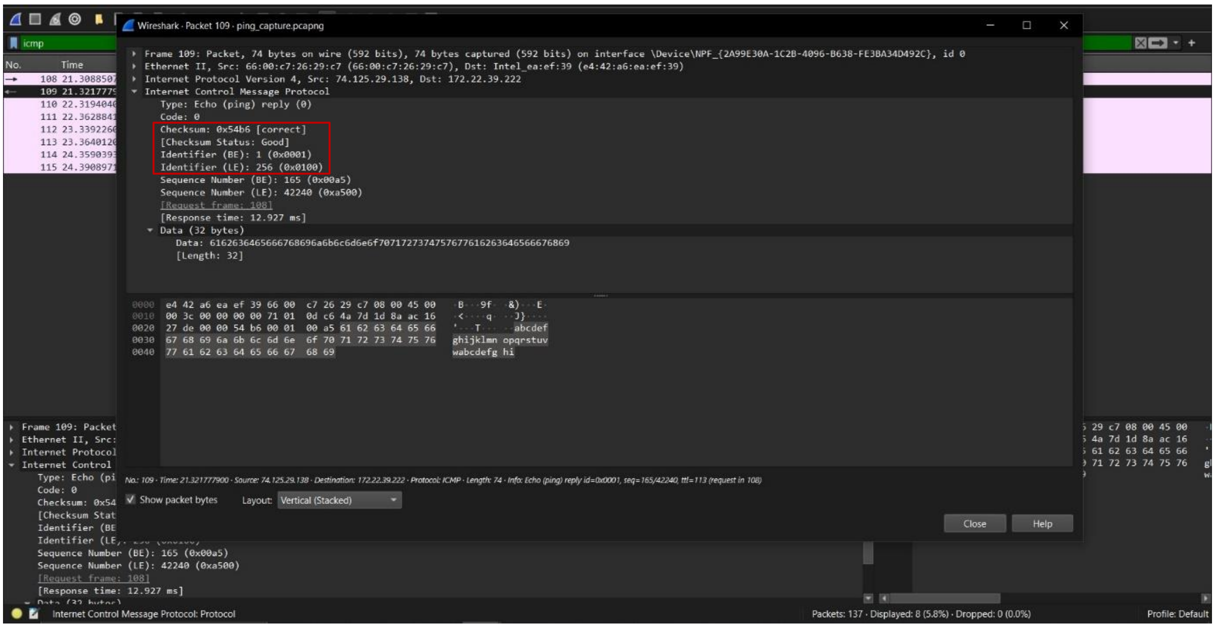


Question 2e: Checksum and Identifier Field Sizes

Checksum: 2 bytes

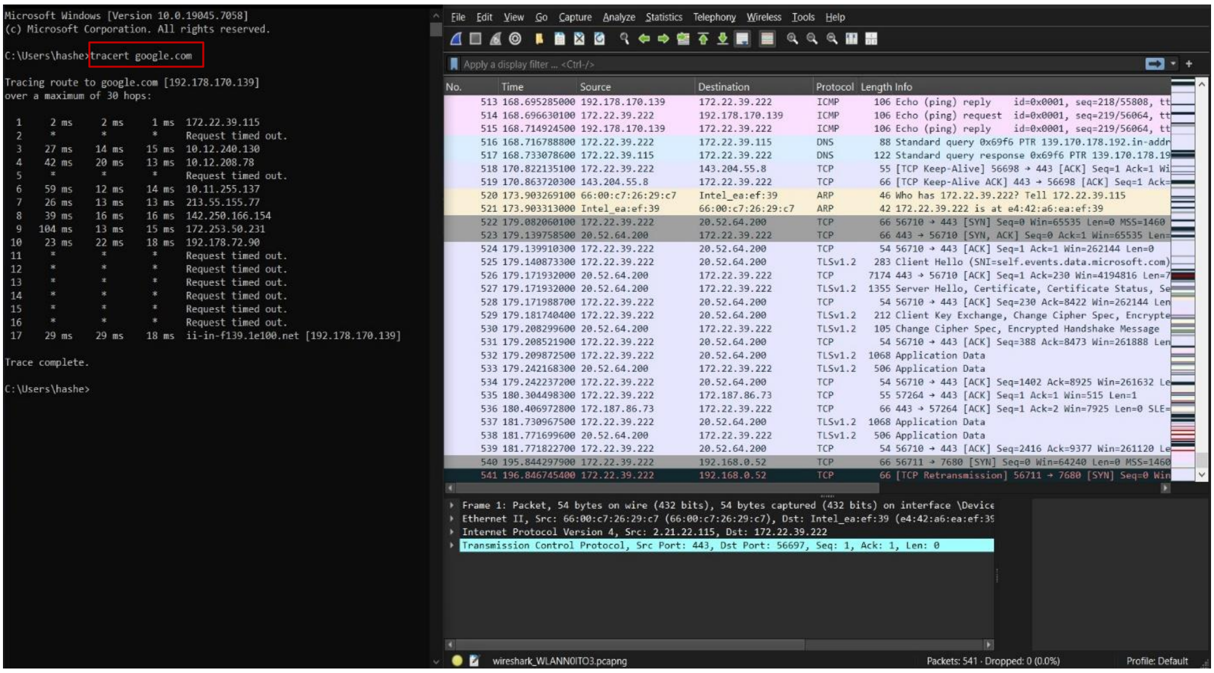
Identifier: 2 bytes

Both fields are 2 bytes each. The Identifier value was 1 (0x0001), which is used to match the reply with its original request (Ghosh, 2021).



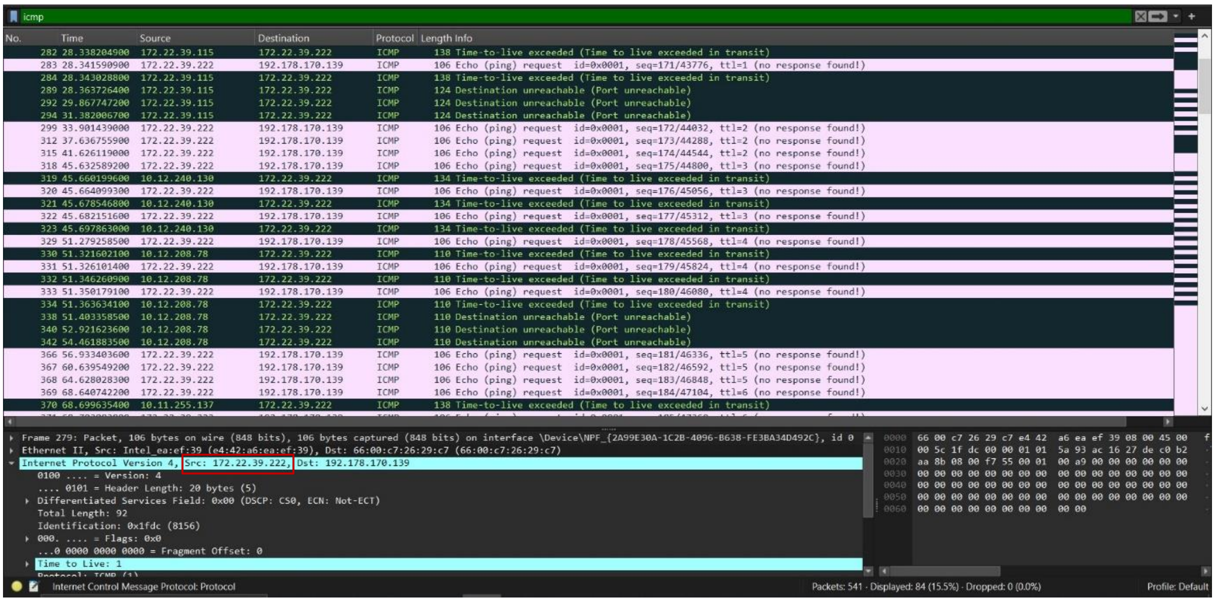
Part 3: Traceroute Analysis

For the second part of the investigation, I ran a traceroute test on google.com using the tracert command. The traceroute showed the complete path from my PC to Google's servers, passing through 17 hops. Some hops showed timeouts which is normal because some routers block ICMP messages for security reasons (Dancuk, 2021).



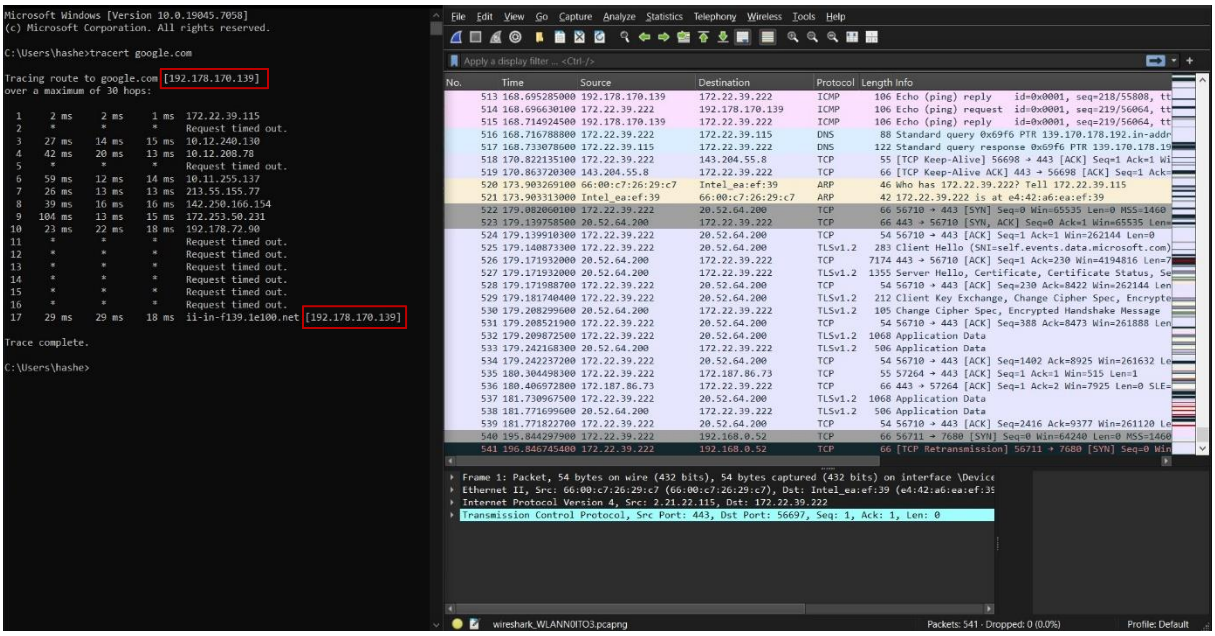
Question 3f: IP Address of Host PC

My PC's IP address as shown in the Wireshark capture and the traceroute results is 172.22.39.222.



Question 3g: IP Address of Target Destination

The IP address of google.com as shown in the traceroute results is 192.178.170.139.



Conclusion

This investigation helped me understand how ICMP packets work in a real network environment. The ping test confirmed active connectivity to google.com with no packet loss, and the traceroute revealed the complete network path. The ICMP request used Type 8 and the reply used Type 0, both with Code 0. All checksums were correct, meaning no packets were corrupted. This kind of analysis is very useful for detecting abnormal ICMP traffic that could indicate attacks such as ICMP flood or ping of death (Ghosh, 2021).

References

Dancuk, M. (2021, August 16). *How to run a traceroute on Linux, Windows and MACOS*. PhoenixNAP. https://phoenixnap.com/kb/how-to-run-traceroute

Ghosh, B. (2021, September 12). *Packet filter analysis for ICMP in Wireshark*. LinuxHint. https://linuxhint.com/pack_filter_icmp_wireshark/

Noviantika, G. (2023, January 4). *How to ping an IP on Windows, MACOS, and Linux*. Hostinger Tutorials. https://www.hostinger.com/tutorials/ping-an-ip